



THE FIRST VACCINE

Jenner noticed that milkmaids infected with cowpox—a similar disease to smallpox, but much milder—didn't catch smallpox. So, he infected a boy with cowpox, and then tried to infect him with smallpox. The boy didn't catch the disease, confirming Jenner's belief that the boy was now immune to smallpox. Seen above is an illustration of Jenner vaccinating his son.

Rabies vaccine being injected into a patient's stomach



Vaccinations

A vaccine is made of weakened forms of the microbes that cause a disease. This is then given to someone (usually as an injection) so their body can prepare itself against the full disease. The first effective vaccine was developed by the British doctor Edward Jenner in the late 18th century to combat smallpox. Since then, scientists have created vaccines for many other diseases.



Syringe with a spare needle and metal case, late 19th century

THE SYRINGE

The invention of the hypodermic syringe in 1853 made it easier for doctors to give vaccinations. It was the work of three men: Irishman Francis Rynd who invented the hollow needle; and French surgeon Charles Pravaz and Scottish physician Alexander Wood who, working independently, devised a way of attaching the needle to a plunger device.

PASTEUR'S WORK

For a while, smallpox was the only disease that people could be vaccinated against. In the early 19th century, scientists didn't know of any other disease with a naturally weak strain of the disease-causing microbe. A breakthrough was made when the French scientist Louis Pasteur (see pp.244–245) discovered how to artificially weaken bacteria, using heat and chemicals, in 1862. This enabled him to produce the first vaccine for rabies in 1885.

◀ TAKING PRECAUTIONS

Here, Pasteur is shown vaccinating people against rabies. He also created a vaccine against anthrax.